Siyu Zou (邹思宇)

Lecturer, Ph.D.
School of Chemical Engineering, Xiangtan University, China
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https://zousiyu1995.github.io/zousiyu/
Born in Songzi (松滋), Hubei, China

Research Interests

My main research interests lie in mathematical modeling and numerical simulation of transport phenomena (fluid dynamics, mass and heat transfer, and reaction kinetics) in nature and industry. I leverage chemical engineering principles to develop multi-physics models to understand, design, and optimize these processes. Some examples of application,

- Modeling and design of biosensor, e.g, enzyme electrodes.
- Modeling and optimization of chemical reactors, e.g, fixed bed reactors, CVD reactors.
- Understanding biological systems from chemical engineering perspective, e.g, eyelashes, digestive system, cardiovascular system.

Employment

Since 09/2024 — **Lecturer**
School of Chemical Engineering, Xiangtan University, Xiangtan, Hunan Province, China

07/2021 – 06/2024 — **Post-Doc in Physical Chemistry**
College of Chemistry, Chemical Engineering and Materials Science, Soochow University, Suzhou, Jiangsu Province, China
Co-Advisors: Prof. Xinjian Feng and Prof. Jie Xiao
Project: “Mathematical Modeling of Three-Phase Interface Enzyme Electrode”

Education

09/2016 – 06/2021 **Ph.D. in Applied Chemistry**
School of Chemical and Environmental Engineering, Soochow University, Suzhou, Jiangsu Province, China
Advisors: Prof. Xiao Dong Chen and Prof. Jie Xiao
Dissertation: "Modeling and Simulation of Reactors with Unique Structures" (Defense 02/06/2021)

09/2016 – 09/2018 **M.Eng. in Chemical Engineering**
School of Chemical and Environmental Engineering, Soochow University, Suzhou, Jiangsu Province, China
Advisor: Prof. Jie Xiao
Successive postgraduate and doctoral programs of study. Transferred to Ph.D. in Applied Chemistry

09/2012 – 06/2016 **B.Eng. in Chemical Engineering**
School of Chemical Engineering and Pharmacy, Jingchu University of Technology, Jingmen, Hubei Province, China

Skills

Expertise Chemical Engineering, Transport Phenomena, Incompressible Flows, Mass Transfer, Heat Transfer, Mathematical Modeling, Numerical Simulation, Multi-Physics Modeling, Computational Fluid Dynamics (CFD), Finite Volume Method (FVM), Turbulence Modeling, Porous Media

Software Basilisk, OpenFOAM, Blender, LaTeX, Linux, ANSYS Fluent, COMSOL Multiphysics

Programming Python, C, MATLAB, Algorithms, Data Structures

Languages Chinese (native speaker), English (intermediate)

Honours and Awards

01/12/2024 **Best Oral Presentation Award**: In the 4th National Conference on Process Modeling and Simulation (hosted by South China University of Technology), my talk "Mathematical Modeling of Three-Phase Interface Enzyme Electrode" was honored with the Best Oral Presentation Award. This work has been published in AIChE Journal and Industrial & Engineering Chemistry Research.

07/2022 **Jiangsu Funding Program for Excellent Postdoctoral Talent**: This funding is provided by the Jiangsu provincial government.

16/07/2020 **Best Student Oral Presentation Award**: In the 2nd National Conference on Process Modeling and Simulation, my talk on eyelashes ("Inhibition of Ocular Water Evaporation by Eyelashes: Computer Simulation and Mechanism Analysis") was honored with the Best Student Oral Presentation Award (1 of 12 awardees out of 62 student talks). This work has been published in Journal of the Royal Society Interface. This conference was organized by the Simulation & Virtual Process Engineering Committee, one of the professional committees of the Chemical Industry and Engineering Society of China (CIESC).

Publications

∇ denotes the corresponding author. Also find me on Google Scholar, Research Gate, and Github.

Main Publications

7. **Siyu Zou**, Jie Xiao∇, and Xinjian Feng∇. "Engineering Enzyme Electrode with 3D Three-Phase Interface to Boost Enzymatic and Electrochemical Cascade Reactions". In: *Chemical Engineering Science* 318 (Dec. 2025), p. 122189. ISSN: 0009-2509. DOI: `10.1016/j.ces.2025.122189`
6. **Siyu Zou**∇, Jie Xiao, and Xiao Dong Chen∇. "A Comprehensive Comparison of Different Reynolds-Averaged Navier–Stokes Turbulence Models in Modeling Turbulent Plane Jets". In: *ACS Omega* 10.19 (May 20, 2025), pp. 19873–19886. ISSN: 2470-1343. DOI: `10.1021/acsomega.5c01448`
5. **Siyu Zou**, Jie Xiao∇, and Xinjian Feng∇. "Modeling Enzymatic and Electrochemical Cascade Reactions at the Three-Phase Interface Enzyme Electrode". In: *AIChE Journal* 70.6 (Mar. 2024), e18420. ISSN: 1547-5905. DOI: `10.1002/aic.18420`
4. **Siyu Zou**, Dandan Wang, Jie Xiao∇, and Xinjian Feng∇. "Mathematical Model for a Three-Phase Enzymatic Reaction System". In: *Industrial & Engineering Chemistry Research* 62.10 (Mar. 2023), pp. 4337–4343. ISSN: 1520-5045. DOI: `10.1021/acs.iecr.2c04492`
3. **Siyu Zou**, Jie Xiao∇, Viola Wu, and Xiao Dong Chen∇. "Analyzing Industrial CVD Reactors Using a Porous Media Approach". In: *Chemical Engineering Journal* 415 (July 2021), p. 129038. ISSN: 1385-8947. DOI: `10.1016/j.cej.2021.129038`
2. **Siyu Zou**, Jinping Zha, Jie Xiao∇, and Xiao Dong Chen. "How Eyelashes Can Protect the Eye through Inhibiting Ocular Water Evaporation: A Chemical Engineering Perspective". In: *Journal of The Royal Society Interface* 16.159 (Oct. 2019), p. 20190425. ISSN: 1742-5662. DOI: `10.1098/rsif.2019.0425`
1. **Siyu Zou**, Ersuo Ling, Shurong Le, Shengpeng Sun, Zhangxiong Wu, Xiao Dong Chen, Duo Wu, and Jie Xiao∇. "Numerical Simulation and Analysis of the Catalytic Ozonation Reactor". In: *Chemical Industry and Engineering Progress* 38.9 (May 15, 2019), pp. 3969–3978. ISSN: 1000-6613. DOI: `10.16085/j.issn.1000-6613.2018-2476` (In Chinese)

Other Publications

11. Lihui Huang, **Siyu Zou**, Mengli Zeng, Yaolan Li, Zhiping Liu, Jun Zhang[∇], and Xinjian Feng[∇]. "Efficient Synthesis of Pyruvic Acid from Biomass Based on Gas-Liquid-Solid Triphase Bioelectrochemical Cascade Reaction". In: *ChemSusChem* 19.6 (2026), e202502726. ISSN: 1864-564X. DOI: 10.1002/cssc.202502726
10. Mengli Zeng, **Siyu Zou**, Lihui Huang, Jun Zhang[∇], Jiong Wang[∇], and Xinjian Feng[∇]. "Photothermally Driven Efficient CO_2 Electroreduction Based on a Superhydrophobic Electrode". In: *Journal of the American Chemical Society* 148.4 (Feb. 4, 2026), pp. 4475–4483. ISSN: 0002-7863, 1520-5126. DOI: 10.1021/jacs.5c19088
9. Jing Li, **Siyu Zou**, Xinjian Feng[∇], and Jie Xiao[∇]. "Reaction Intensification in Triphase Enzyme Electrode: In Silico Design of Interfacial Microenvironment". In: *AIChE Journal* (Jan. 7, 2026), e70210. ISSN: 0001-1541, 1547-5905. DOI: 10.1002/aic.70210
8. Tianmin Wang, **Siyu Zou**, Peng Wu, Renpan Deng, Chaoping Fu, Changyong Li[∇], and Ai-Zheng Chen. "CFD Modeling of Flow, Mixing, Digestion, and Absorption in the Small Intestine". In: *Food Engineering Reviews* (Dec. 2025). ISSN: 1866-7910, 1866-7929. DOI: 10.1007/s12393-025-09421-w
7. Zhiping Liu, **Siyu Zou**, Xi Chen, Lihui Huang, Xia Sheng[∇], and Xinjian Feng[∇]. "Interfacial Microenvironment and Catalyst Modulation for Efficient Hydrogen Peroxide Synthesis *via* Mimicking Oxidase Catalysis". In: *Materials Horizons* 13.2 (2026), 10.1039.D5MH01332A. ISSN: 2051-6347, 2051-6355. DOI: 10.1039/D5MH01332A
6. Xi Chen, Xia Sheng[∇], **Siyu Zou**, Zhiping Liu, Jingyu Lu, Zhaoyue Tan, and Xinjian Feng[∇]. "Creation of a Triphasic Reaction Interface to Boost Colloidal Photooxidation under Visible Light". In: *Langmuir* 41.33 (Aug. 26, 2025), pp. 22152–22159. ISSN: 0743-7463, 1520-5827. DOI: 10.1021/acs.langmuir.5c02306
5. Kaixin Li, **Siyu Zou**, Jun Zhang[∇], Yang Huang, Lin He[∇], and Xinjian Feng[∇]. "Superhydrophobicity-Enabled Efficient Electrocatalytic CO_2 Reduction at a High Temperature". In: *ACS Catalysis* 13.14 (June 2023), pp. 9346–9351. ISSN: 2155-5435. DOI: 10.1021/acscatal.3c01444
4. Xiao Dong Chen[∇] and **Siyu Zou**. "Reaction Engineering Approach to Turbulence Modelling—Universal Law of the Wall, Pipe Flow, and Planar Jet Flow". In: *Journal of Chemical Engineering of Japan* 54.1 (Jan. 2021), pp. 1–11. ISSN: 1881-1299. DOI: 10.1252/jcej.20we056
3. Jinping Zha, **Siyu Zou**, Jianyu Hao, Xinjuan Liu, Guillaume Delaplace, Romain Jeantet, Didier Dupont, Peng Wu, Xiao Dong Chen, and Jie Xiao[∇]. "The Role of Circular Folds in Mixing Intensification in the Small Intestine: A Numerical Study". In: *Chemical Engineering Science* 229 (Jan. 2021), p. 116079. ISSN: 0009-2509. DOI: 10.1016/j.ces.2020.116079
2. Hongtao Xia, **Siyu Zou**, and Jie Xiao[∇]. "Numerical Simulation of Shear-Thinning Droplet Impacting on Randomly Rough Surfaces". In: *CIESC Journal* 70.2 (Feb. 5, 2019), pp. 634–645. ISSN: 0438-1157. DOI: 10.11949/j.issn.0438-1157.20181213 (In Chinese)

1. Jie Xiao$^{\nabla}$, Fei Pan, Hongtao Xia, **Siyu Zou**, Hui Zhang, Oluwafemi Ayodele George, Fei Zhou, and Yin-lun Huang. “Computational Study of Single Droplet Deposition on Randomly Rough Surfaces: Surface Morphological Effect on Droplet Impact Dynamics”. In: *Industrial & Engineering Chemistry Research* 57.22 (May 2018), pp. 7664–7675. ISSN: 1520-5045. DOI: 10.1021/acs.iecr.8b00418

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